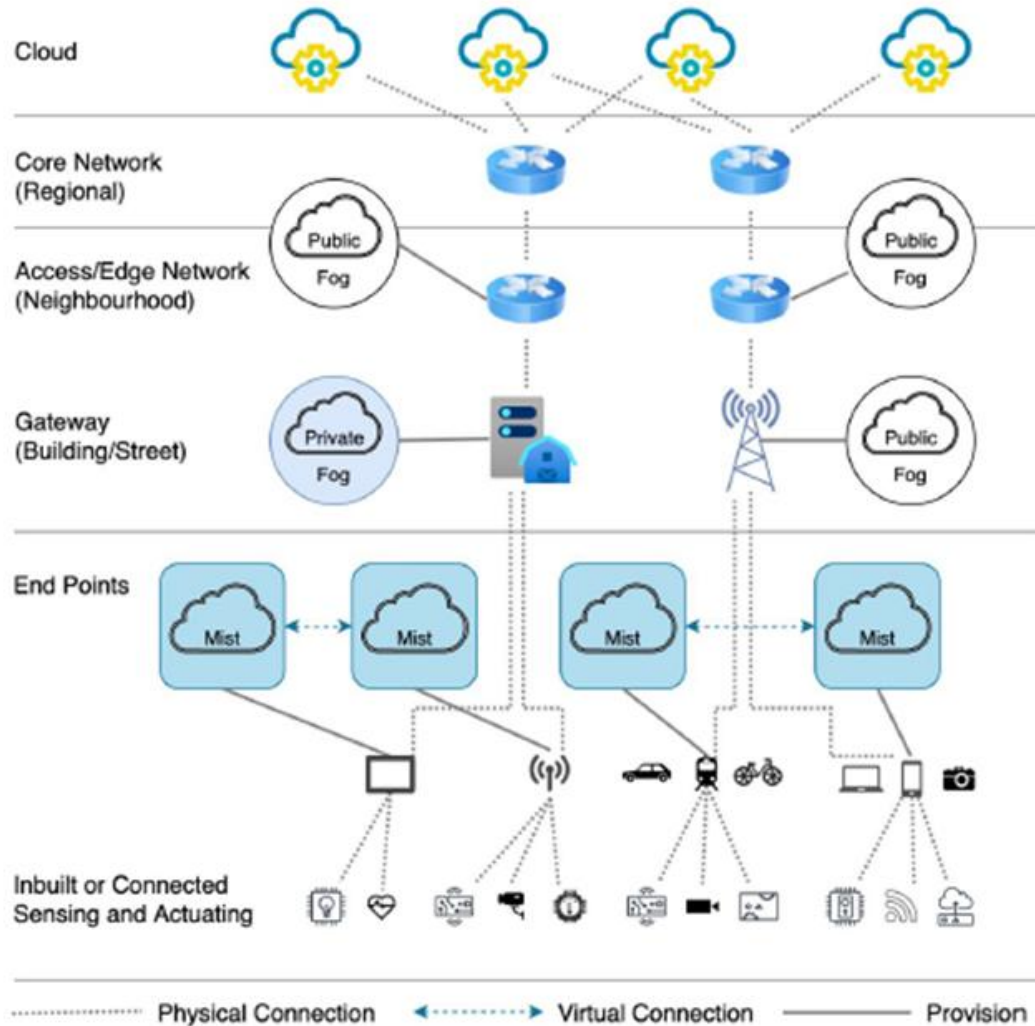
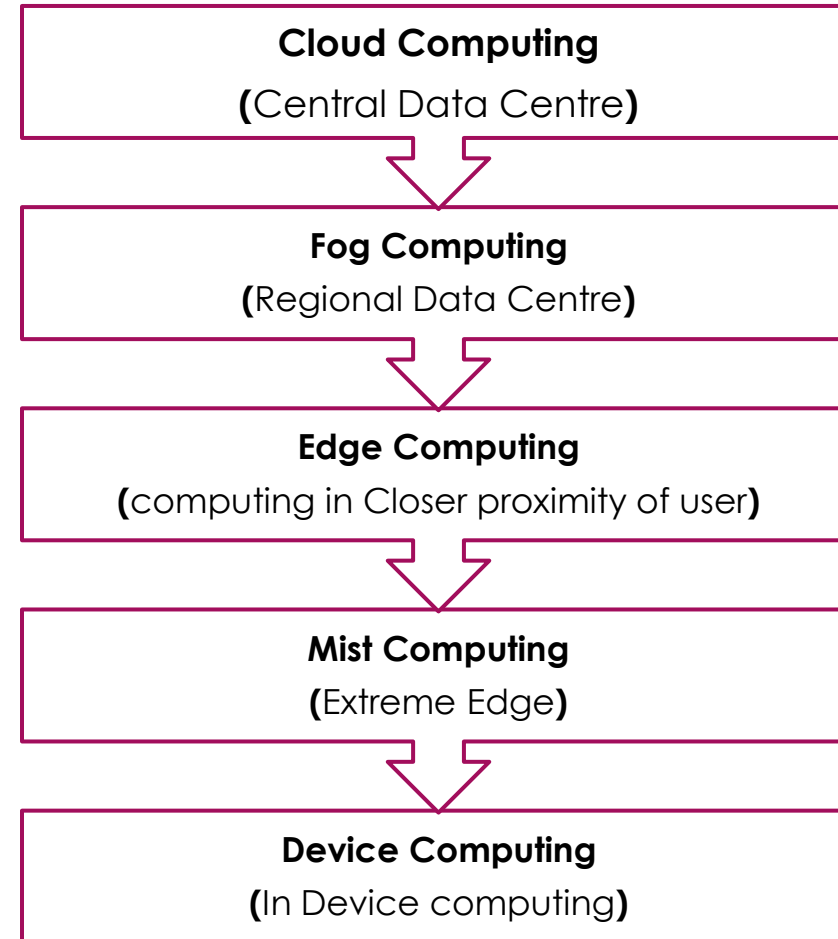


IoT Multilayered Architecture - recap



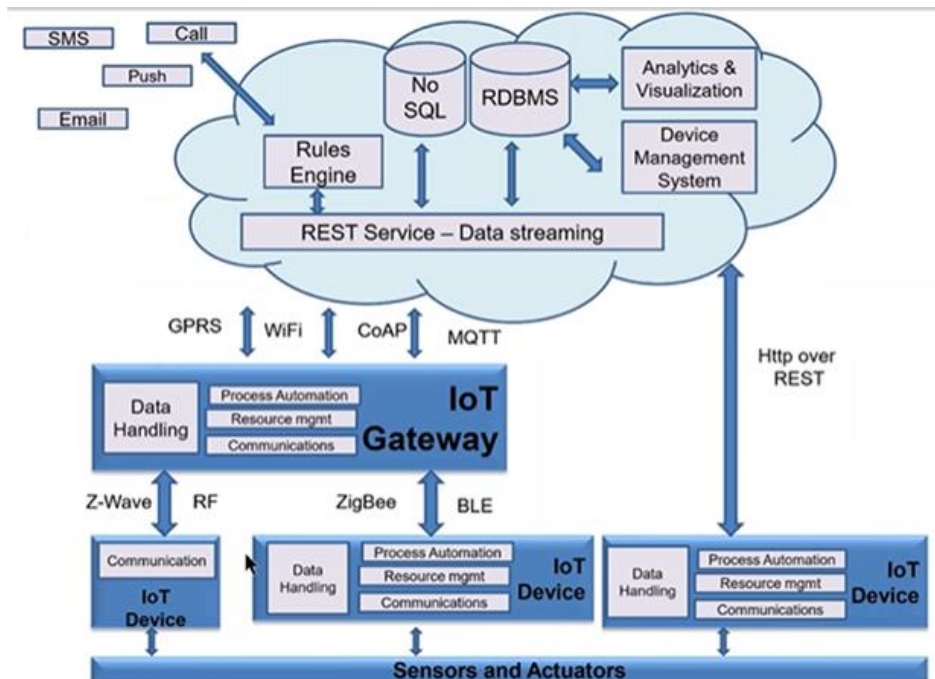
Data processing happens at multiple layers



What is Cloud Computing in IoT?

Conceptual Definition

- Centralized computing and storage platform
- Always available and Highly scalable
- Integrates with enterprise IT systems



Role of Cloud in IoT



**DATA
INGESTION**



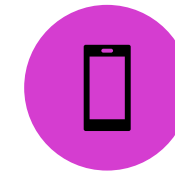
**LONG-TERM
STORAGE**



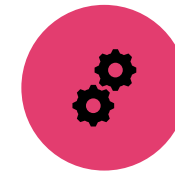
**ANALYTICS &
DASHBOARDS**



**AI/ML MODEL
TRAINING**



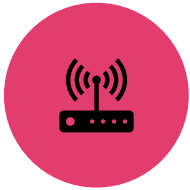
**DEVICE
MANAGEMENT**



**INTEGRATION
WITH BUSINESS
SYSTEMS**

Typical IoT Cloud Components

Typical IoT Cloud Components: Core Building Blocks



IOT INGESTION
SERVICE



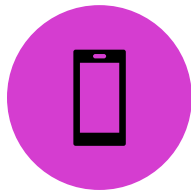
MESSAGE
BROKERS



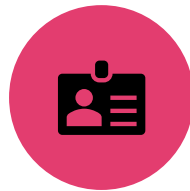
DATA STORAGE
(STREAM
PROCESSING)



ANALYTICS &
VISUALIZATION



DEVICE
REGISTRY &
MANAGEMENT



SECURITY &
IDENTITY
SERVICES

Limitations of Cloud-Only IoT Architecture



High latency for real-time control



Dependency on internet connectivity



High data transmission costs



Privacy & data residency issues



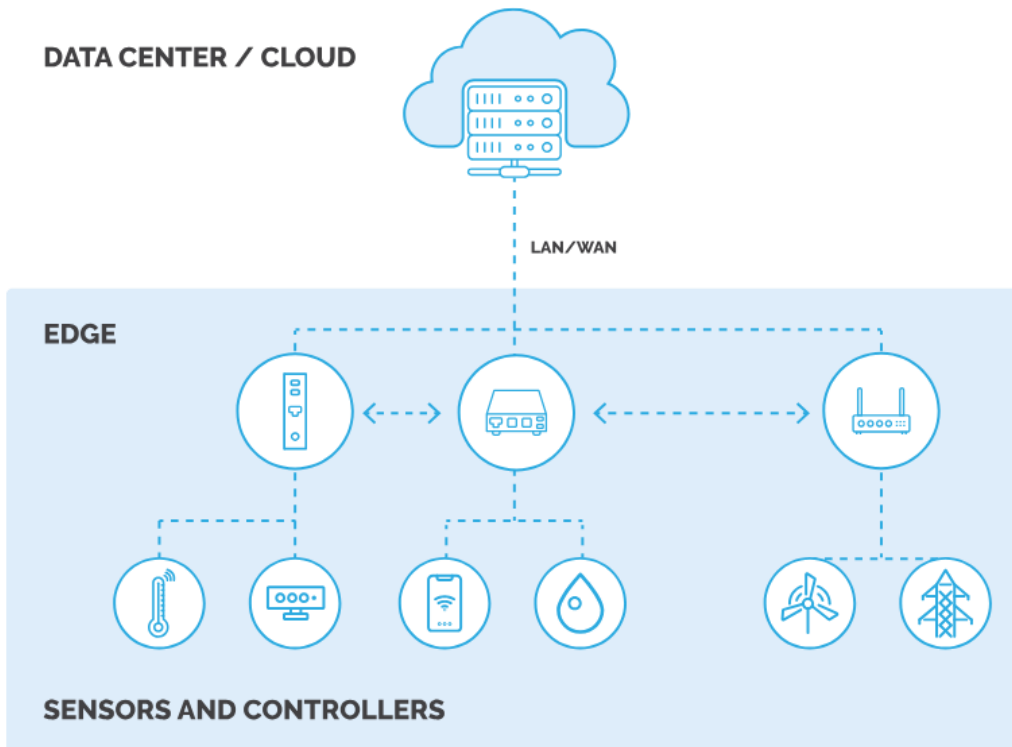
Not suitable for safety-critical systems

This naturally leads to **edge computing**

What is Edge Computing in IoT?

Conceptual Definition

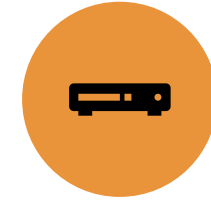
- Processing data **closer to the source**
- Happens near or at the device
- Reduces dependency on cloud



IOT DEVICE ITSELF



EDGE GATEWAY



LOCAL SERVER /
INDUSTRIAL
CONTROLLER

What Happens at the Edge

- Data filtering & aggregation
- Threshold-based decisions
- Local rule execution
- Protocol translation
- Temporary storage
- Fail-safe operations
- Edge AI inference

Edge is about **speed, autonomy, and resilience**

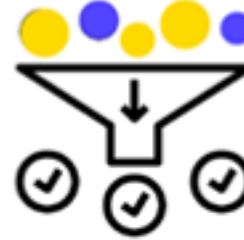
Data Flow in an IoT System



**Sensors generate
raw data**



**Device
preprocesses
data**



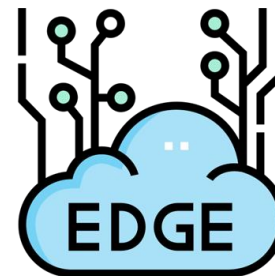
**Edge filters &
applies logic**



**Important data
sent to cloud**

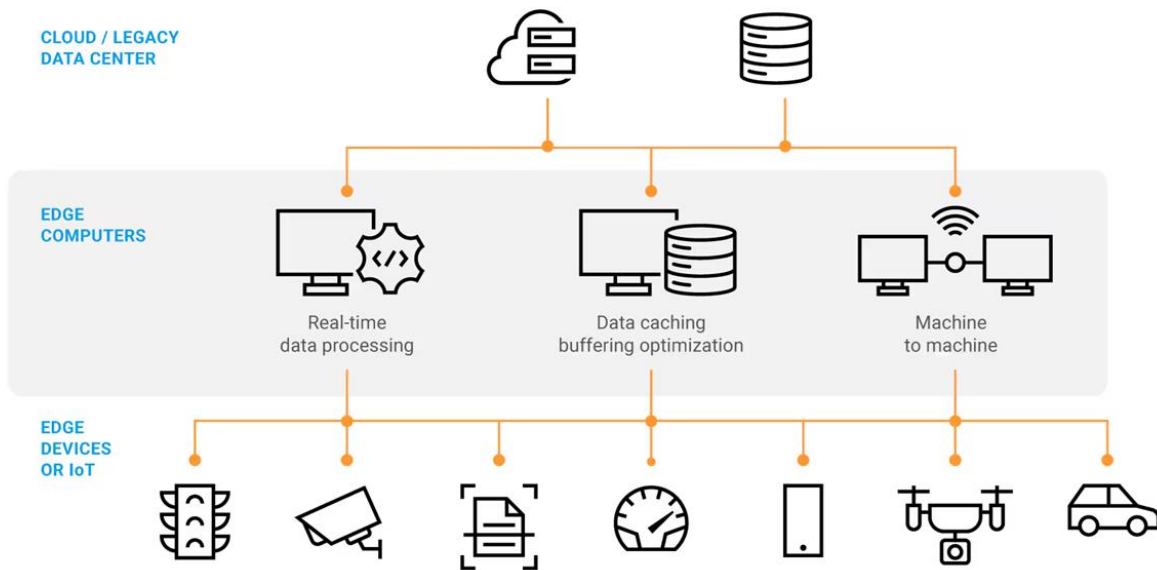


**Cloud performs
analytics &
visualization**



**Commands sent
back to
edge/device**

Device vs Edge vs Cloud – Clear Separation



Edge vs Cloud – Decision Criteria

Use Edge When

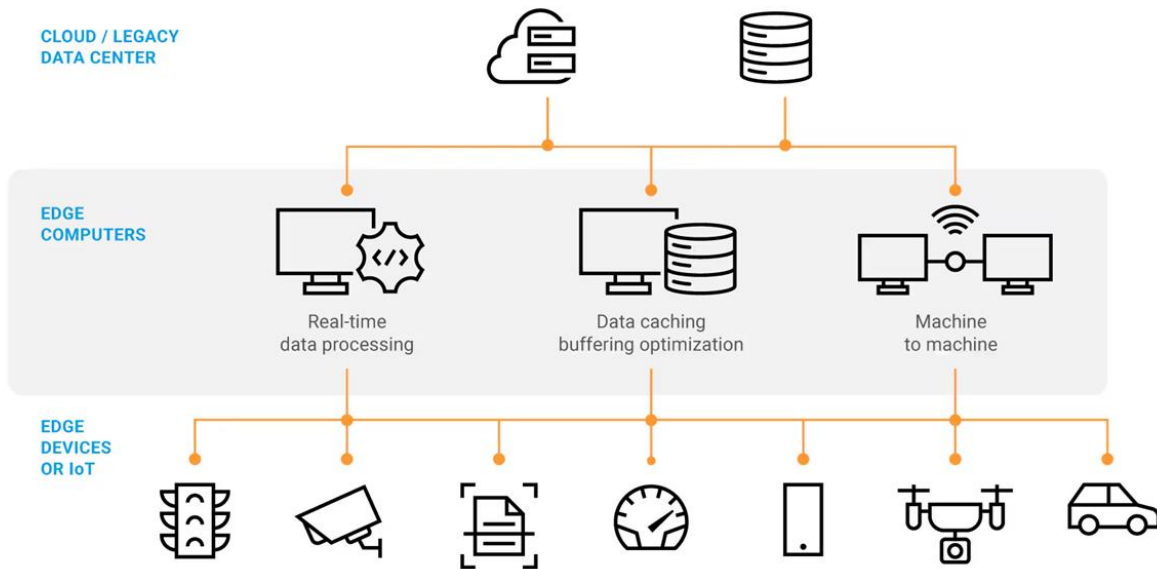
- Low latency is required
- Connectivity is unreliable
- Safety-critical decisions are needed
- Bandwidth is limited

Use Cloud When

- Long-term analytics needed
- Cross-site correlation required
- AI/ML model training
- Business reporting & dashboards

Layer	Primary Role	Example
Device	Sense & Act	Sensor controller
Edge	Local intelligence	Gateway / edge server
Cloud	Global intelligence	Analytics & dashboards

Real-World IoT Use Case Mapping



Smart Factory

- **Edge:** machine monitoring, anomaly detection
- **Cloud:** predictive maintenance analytics

Smart City

- **Edge:** traffic signal control
- **Cloud:** traffic optimization & planning

Healthcare

- **Edge:** real-time patient alerts
- **Cloud:** patient history & analytics

Edge Intelligence (AI at the Edge)

Edge Intelligence, often called **AI at the Edge**, is the ability to run artificial intelligence models directly on edge devices or gateways instead of relying solely on the cloud.



Faster response



Reduced data transfer



Better privacy



Works offline



**IMAGE
RECOGNITION
AT EDGE**



**VIBRATION
ANOMALY
DETECTION**



**PREDICTIVE
ALERTS**

Edge Intelligence brings speed, autonomy, and resilience to IoT systems, making them smarter and more reliable.

Security Responsibilities – Edge vs Cloud



Edge Security

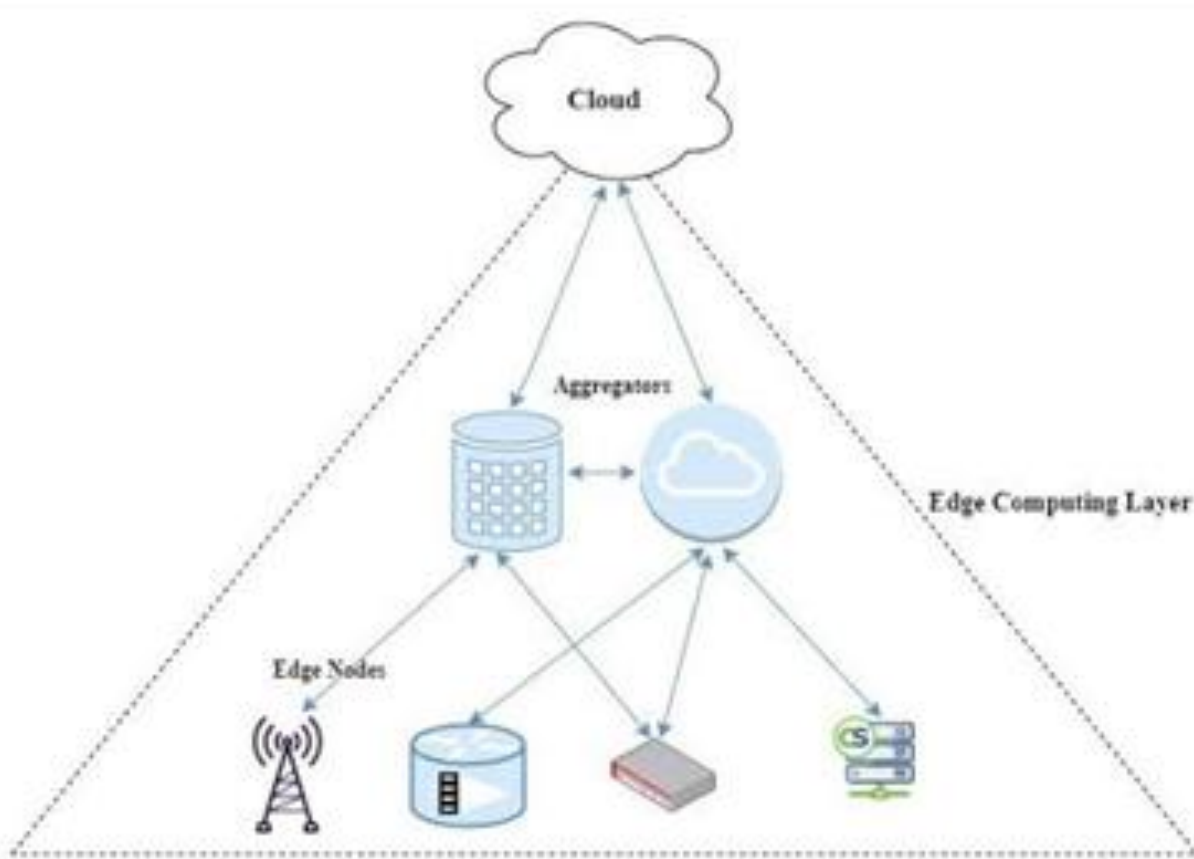
- Device authentication
- Secure communication
- Local access control
- Tamper resistance



Cloud Security

- Identity management
- Data encryption
- Role-based access
- Compliance & auditing

Common Mistakes in Cloud & Edge Design



Sending all raw data to cloud

Ignoring edge intelligence

Overcomplicating edge logic

Weak security at edge

No fallback strategy during outages

Session Summary – Key Takeaways

Cloud and edge **complement**, not replace each other

Edge enables speed, autonomy, and resilience

Cloud enables scale, analytics, and integration

Smart IoT systems balance **device + edge + cloud**